

Engineering Mechanics: ME101

Lecture 9 & 10 : SFD & BMD



Pankaj Biswas (PhD)
Department of Mechanical Engineering
IIT Guwahati
D Block : Room No 207 : Tel: 2675

Beams

What is a beam?

- A beam is a structural element that is subjected to transverse loads or moments which mainly cause it to bend in the axial plane (i.e., the plane containing the beam axis).
- Beams are suitably supported so as to be in equilibrium under the action of external loads or moments.



Figure 1

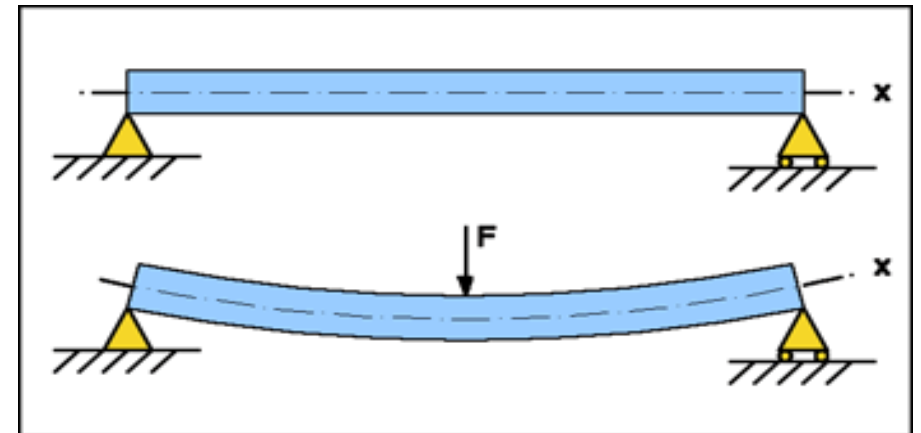
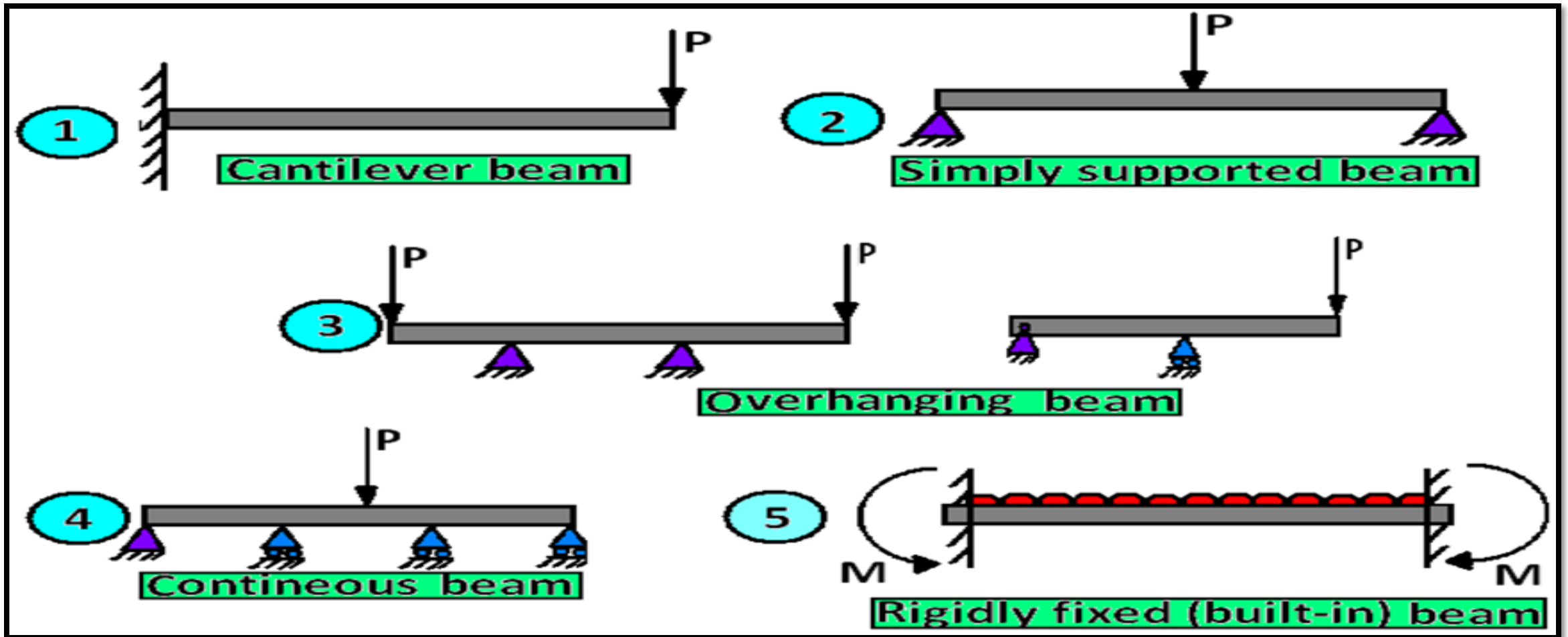


Figure 2

Types of beam

- **Beam:** When the member resist the forces applied laterally or transversely to their axis. Then the member is called a 'beam'.



Concept of SF & BM

- ❖ For a beam, **three force components** are internally developed.
 - Axial force
 - Shear
 - Bending Moment

- ❖ **Shear force:** The SF at the c/s of a beam may be **defined as unbalanced vertical force to the right or left of the c/s.**

- ❖ **Bending moment:** The BM at the c/s of a beam may be defined as algebraic sum of the moments of the force, to the right or left of the c/s.

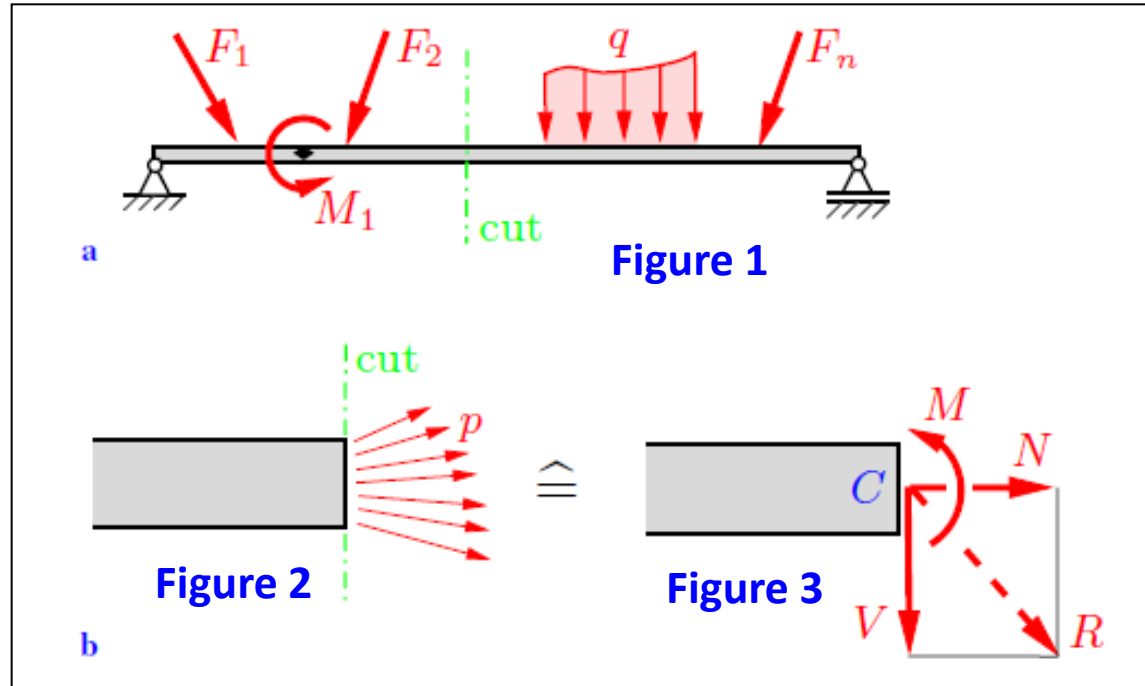
- (**Note:** While calculating SF or BM, **the end reaction must also be considered** along with other external loads.)

Why to know SFD & BMD?

❖ Why to know SFD & BMD?

- ✓ To know the manner in which values of SF & BM changes, plotting of SF & BM is done as ordinate and position of c/s as abscissa.
- ✓ These diagrams are very useful as they give a clear picture of the distribution of SF & BM along the beam.
- ✓ Magnitude and location of different quantities can be easily visualized.
- ✓ SFD & BMD are essential for designers to make decisions on the shape and size of a beam.
- ✓ To compute the area of the cross-section required to sustain a given load.
- ✓ To compute the deformation.

Stress Resultants



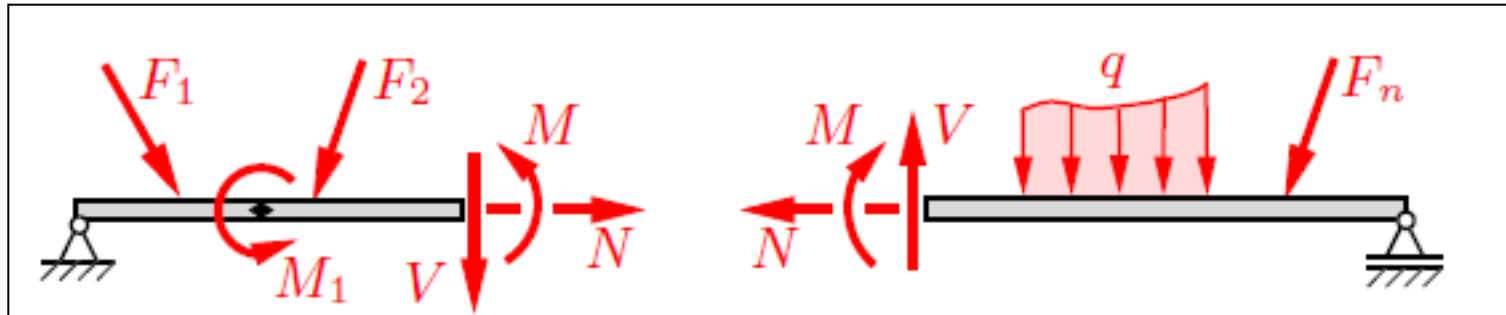
❑ The quantities N , V and M are called the stress resultants.

❑ In particular,

- ✓ N is called the normal force,
- ✓ V is the shear force and
- ✓ M is the bending moment.

Analysis of Stress Resultants

- ❖ In order to determine the **stress resultants**, the beam may be **divided by a cut into two segments (method of sections)**. A free body diagram of each part of the beam will include all of the forces acting on the respective part, i.e., the applied loads (forces and couples), the support reactions and the stress resultants acting at the cut sections.



Sign Convention of BM & SF

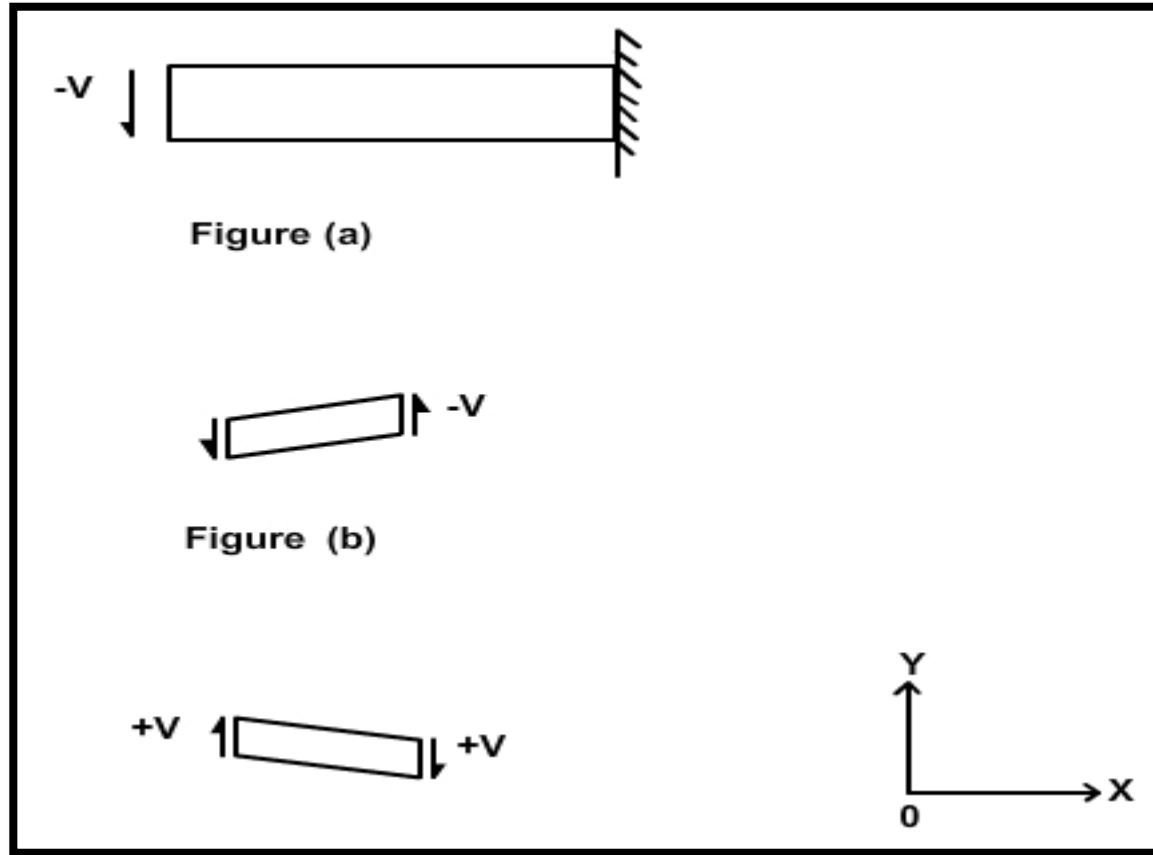


Figure (1)

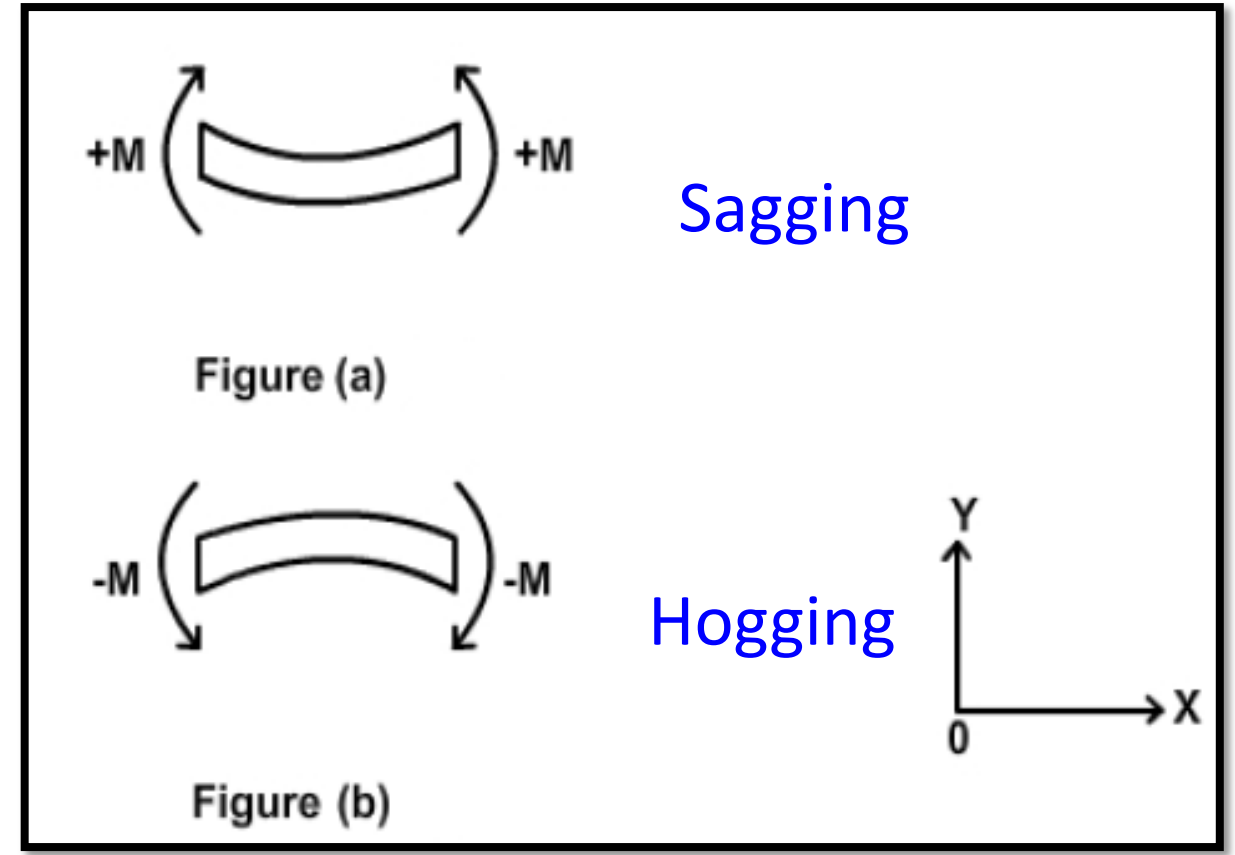


Figure (2)

Procedures for Finding out SF and BM

- Direct Method/Method of Sections**
- Beams Element method: Differential equations of Equilibrium**

Direct Method/Method of Sections

❖ **This method is illustrated by the following example:**

- ✓ For the cantilever as shown in figure below subjected to uniformly distributed load.

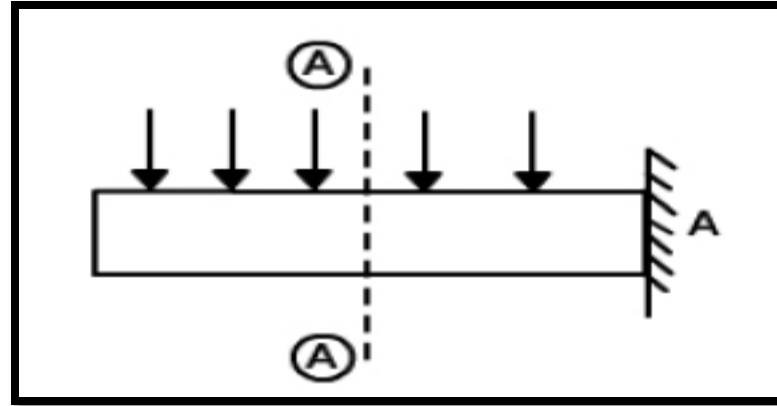


Figure (1)

- ✓ Cut the cantilever along section A-A and obtain free body diagram as given in below figure.

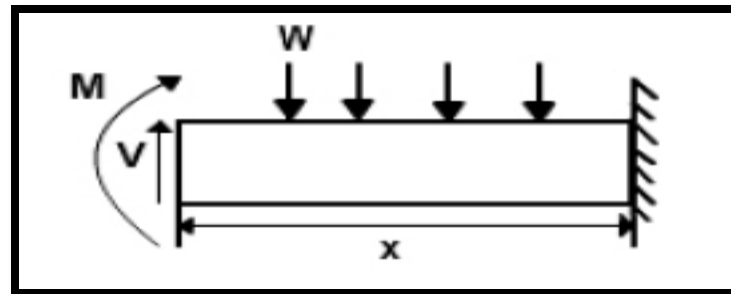


Figure (2)

- ✓ The segment of the beam shown in above figure is in equilibrium under the action of external forces and internal forces and moments.

Beams Element Method: Differential Equations of Equilibrium

❖ This method is illustrated by the following example:

✓ For the cantilever as shown in figure below subjected to uniformly distributed load.

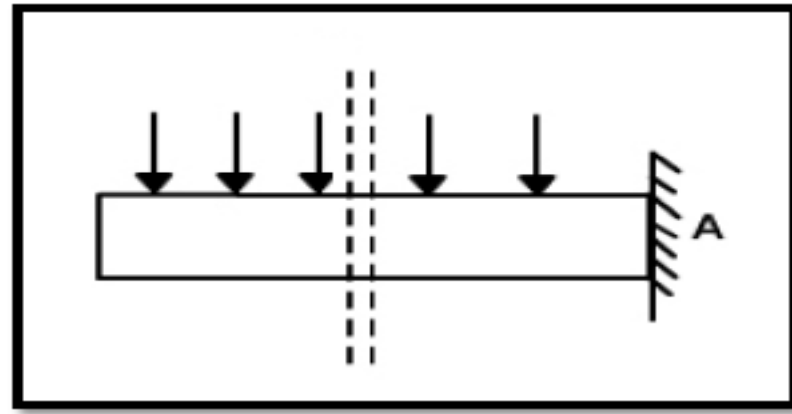
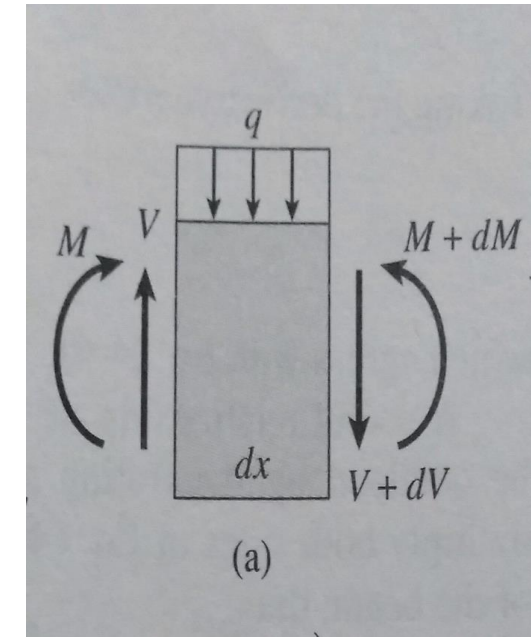


Figure (1)



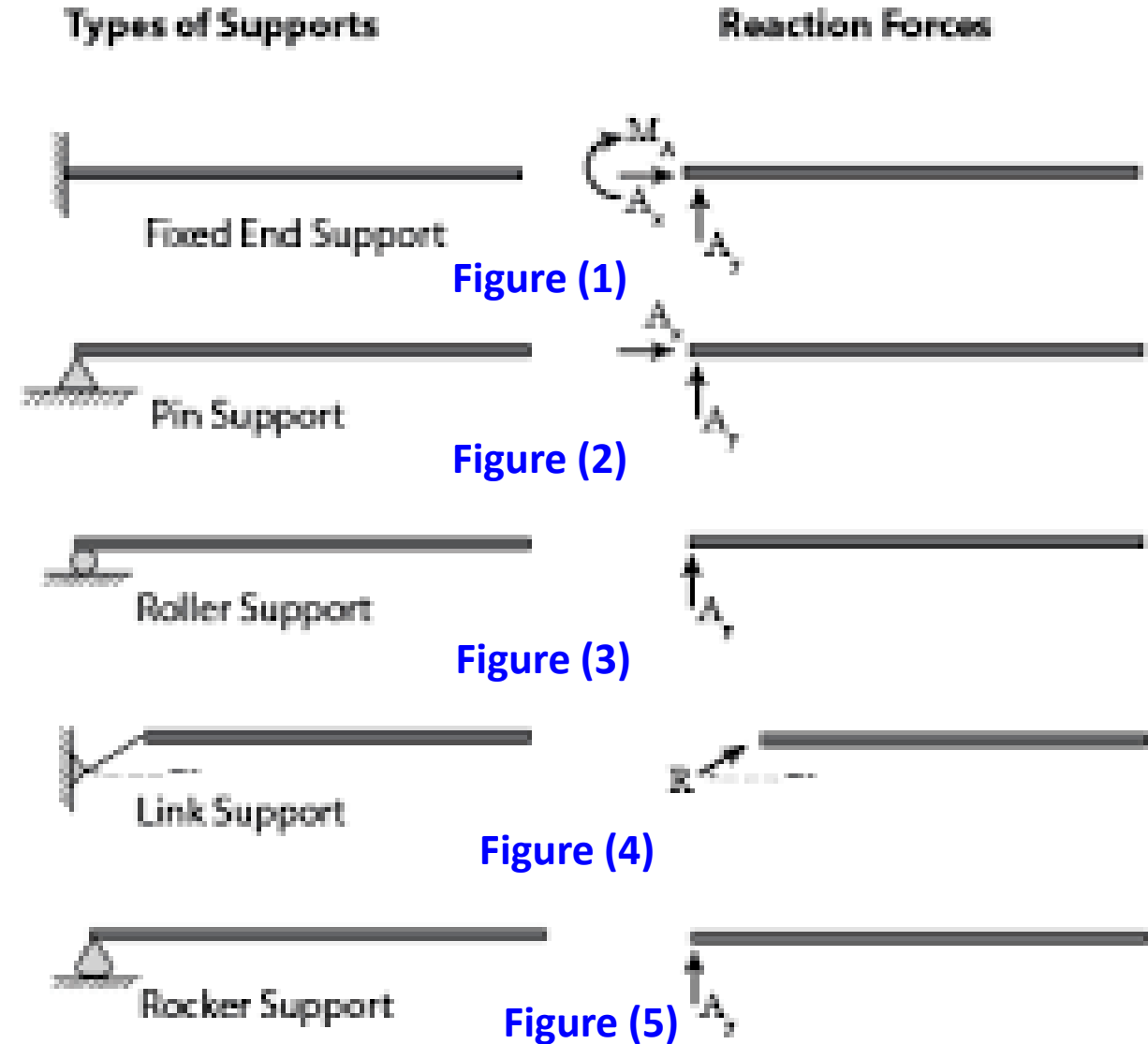
Important points to be remembered

Type of loading	SFD	BMD
1. Point load	Vertically sudden changes	Remain same
2. No load between two points	SF does not changes (SF line is horizontal)	BM changes linearly (inclined straight line)
3.UDL between two points	SF changes linearly (inclined straight line)	BM changes in parabolic law (parabolic line)
4.UVL (Uniformly varying load) between two points	SF changes in parabolic law (parabolic line)	BM changes in cubic law (cubic line)

Types of supports

In this section, different types of supports and their reaction forces on the beam have been shown.

- **Fixed support:** Both horizontal and vertical reaction forces and also reaction moment occur.
- **Pin or Hinge support:** Only horizontal and vertical reaction forces occur but no reaction moment is present.
- **Roller support:** One and only vertical reaction force is present.

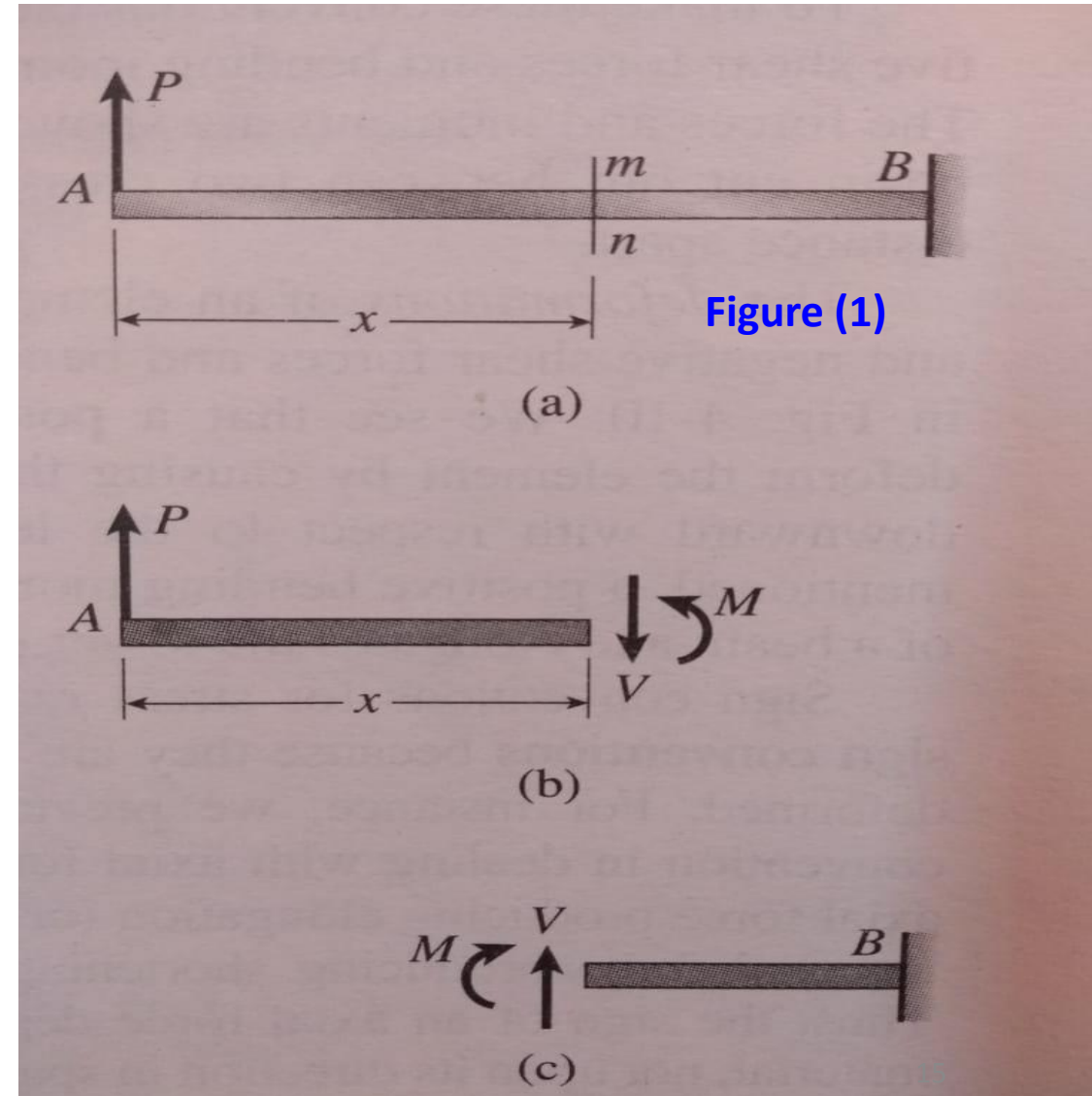


Construction of SF & BM Diagrams

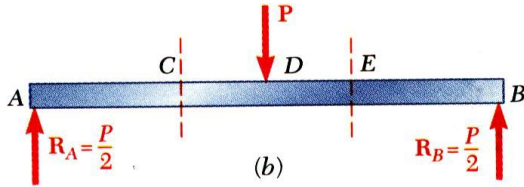
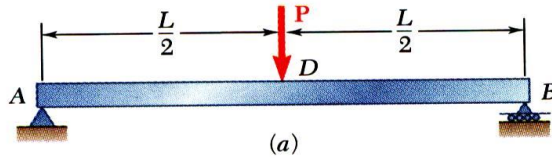
1. Find reactions from loading diagram.
2. Sum up all vertical components of force and reactions with scanning the beam from left to right to preserve the mathematical sign conventions adopted.
3. Shear force=sum of all vertical forces to the left of the section.
4. No shear act through the beam just beyond the last vertical force.
5. Now, for BMD sum up the area of SFD along the length from LHS.
6. Constant SF produce uniform change in BM(straight inclined line) and zero shear produce constant BM.
7. Also, $dM/dx = V$ or $F \rightarrow$ maximum or minimum BM at SF=0.
8. **Validity check (checking terminal condition):**
 - Free or pinned end $\rightarrow$ computed sum of moment=0
 - Built-in end $\rightarrow$ moment computed by summation = one which is calculated initially for reaction

Shear Forces and Bending Moments

1. When a beam is loaded by forces or couples, stresses and strains are developed throughout the interior of the beam.
2. The cantilever beam is subjected to a force **P**. We cut through the beam at a cross section mn located at a distance x from the free end. Isolate the left hand part of the beam as free body and the free body is held in equilibrium by force **P** and the **internal stress and moment**.
3. From statics, we know that the resultant of the stresses acting on the cross-section can be reduced to a **shear force (V)** and a **bending moment (M)**.

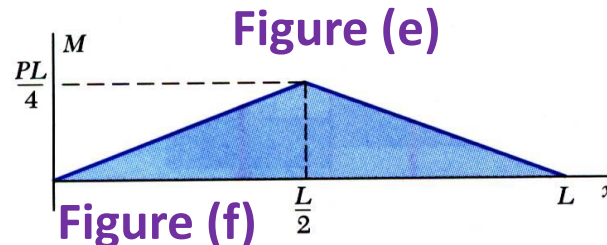
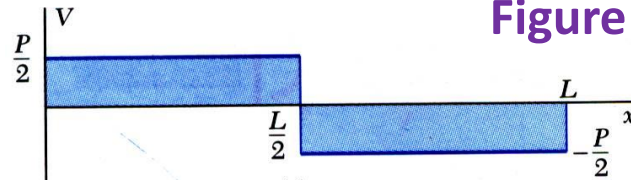
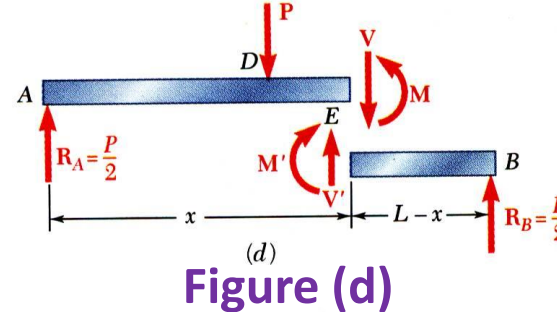
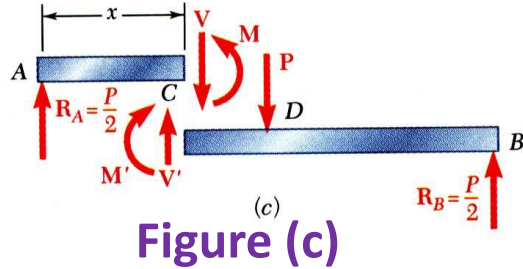
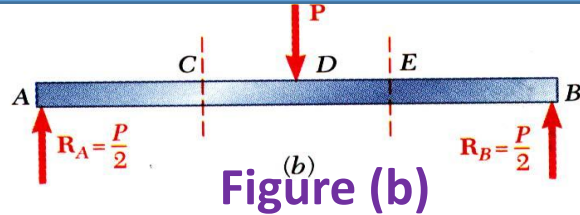


Shear Force and Bending Moment Diagrams



- Draw the Variation of shear and bending moment along beam may be plotted.
- Determine reactions at supports.

Shear Force and Bending Moment Diagrams



1. Determine reactions at supports.

2. Cut beam at C and consider member AC ,

$$V = +P/2 \quad M = +Px/2 \quad (i)$$

3. Cut beam at E and consider member EB ,

$$V = -P/2 \quad M = +P(L-x)/2 \quad (ii)$$

4. For a beam subjected to concentrated loads, shear is constant between loading points and moment varies linearly.

Relationships between Loads, Shear force and Bending Moments

□ For Distributed Loads

1. The rate of change of the shear force at any point on the axis of the beam is equal to the negative of the intensity of the distributed load at that same point (*Note: If the sign convention for q is reversed, then the minus sign is omitted in the below equation*).

$$\sum F_{vertical} = 0$$

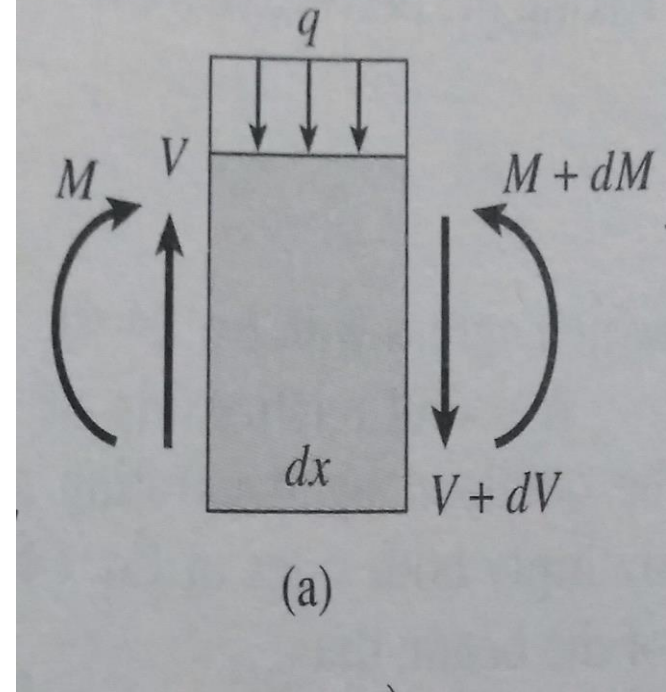
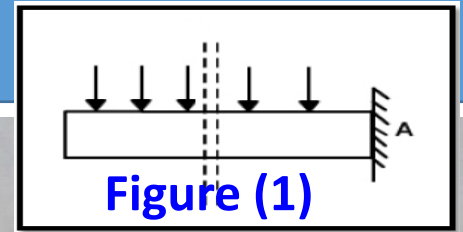
$$V - qdx - (V + dV) = 0 \quad (i)$$

$$\frac{dV}{dx} = -q \quad (ii)$$

2. The rate of change of bending moment at any point on the axis of the beam is equal to the shear force at that same point.

$$\sum M = 0 \Rightarrow -M - qdx\left(\frac{dx}{2}\right) - (V + dV)dx + M + dM = 0 \quad (iii)$$

$$\frac{dM}{dx} = V \quad (iv)$$



Relationship between Loads, Shear Force and Bending Moment

❑ For Concentrated Loads

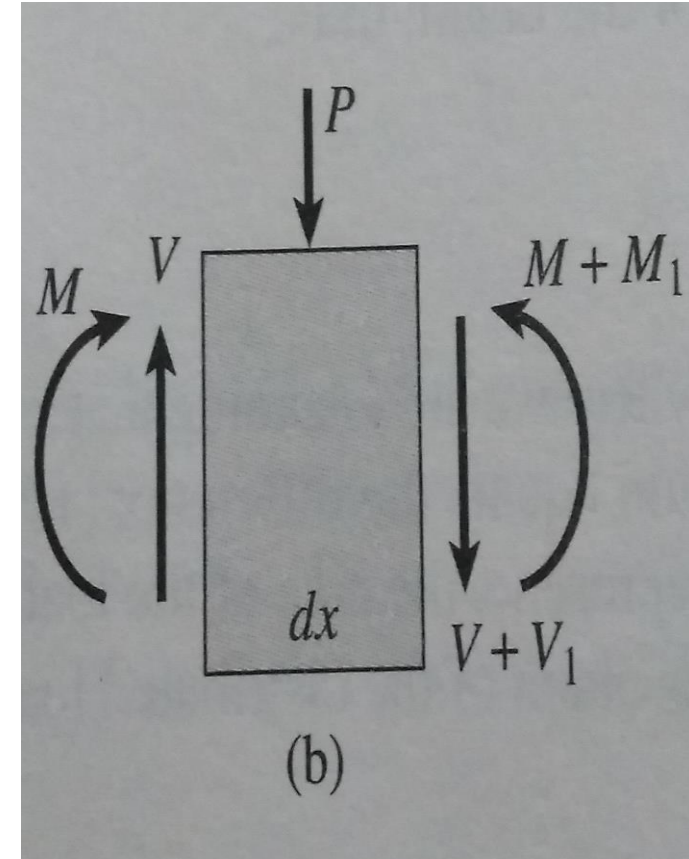
1. Let us consider a concentrated load P acting on the beam element. From equilibrium of forces in the vertical direction, we get

$$V - P - (V + V_1) = 0 \quad (i) \quad \text{or, } V_1 = -P \quad (ii)$$

❖ **Note:** As we pass from left to right through the point of load application, the shear force decreases by an equal amount of P .

- ✓ From equilibrium of moments about the left-hand face of the element, we get

$$-M - P\left(\frac{dx}{2}\right) - (V + V_1)dx + M + M_1 = 0 \quad (iii) \quad \text{or, } M_1 = P\left(\frac{dx}{2}\right) + Vdx + V_1dx$$



Shear Force and Bending Moment Diagrams

Simply Supported Beam with a Single Concentrated Load

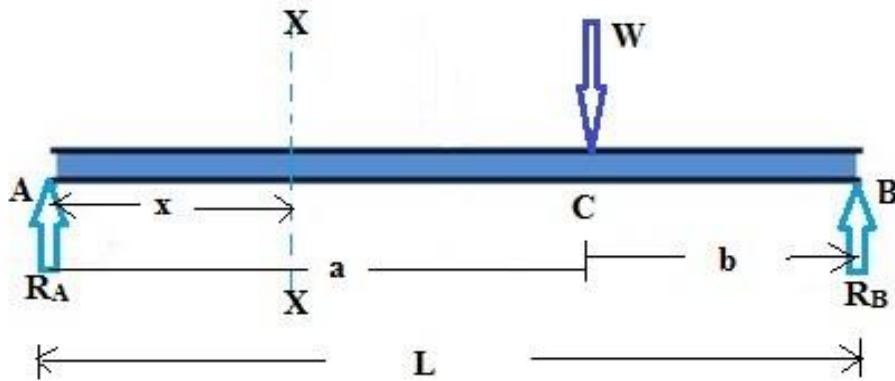


Figure (1)

- Figure shows a single point load W acting at a distance a from the left support of the simply supported beam.
- Two vertical reactions R_A and R_B can be determined from two equilibrium equations.

$$\sum F_{vertical} = 0 \Rightarrow R_A + R_B = W \quad (i)$$

Taking moments of the forces about B and equating to zero i.e., $\sum M_B = 0 \Rightarrow R_A L - Wb = 0$ (ii)

$$R_A = \frac{Wb}{L} \quad \text{and} \quad R_B = \frac{Wa}{L}$$

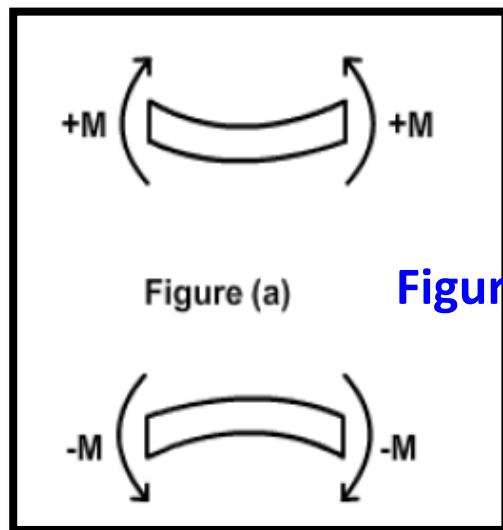
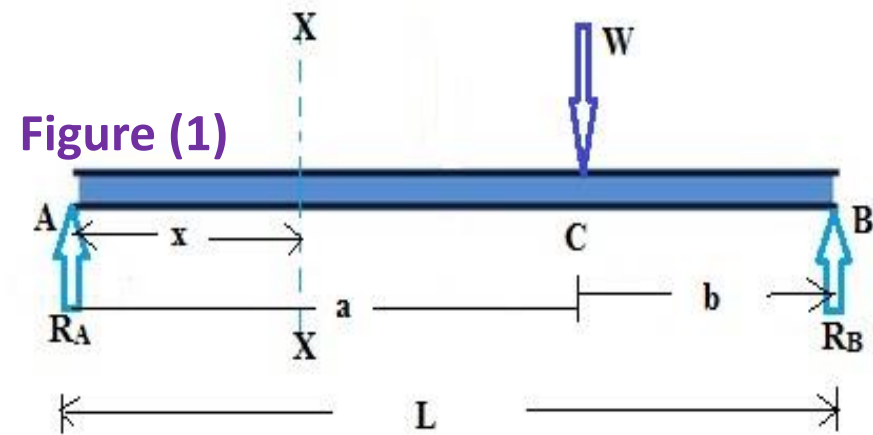


Figure (2)

Figure (1)



➤ For Shear Force Diagram

Consider a section x from point A.

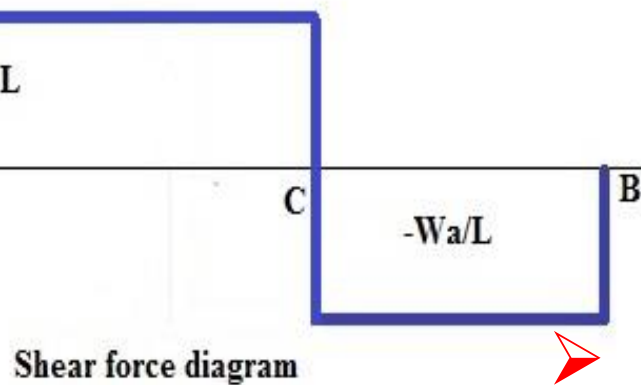
I. Between A and C i.e., $0 \leq x < a$ $V_x = R_A = \frac{Wb}{L}$ (i)

II. At C, shear force changes from $\frac{Wb}{L}$ to $\frac{Wa}{L}$ (ii)

III. Between C and B i.e., $a < x < L$ $V_x = R_A - W = \frac{Wa}{L}$ (iii)

IV. At B, shear force changes from $\frac{Wa}{L}$ to zero (0)

Figure (2)



➤ For Bending Moment Diagram

I. Between A and C ($0 \leq x < a$) $M_x = R_A x = \frac{Wb}{L} x$ (iv)

II. At C, $M_{Cx} = \frac{Wb}{L} a$

III. Between B and C $a < x \leq L$

$$M_x = R_A x - W(x - a) \quad (v)$$

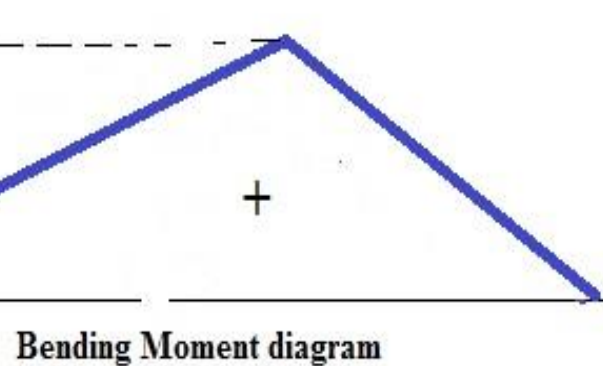
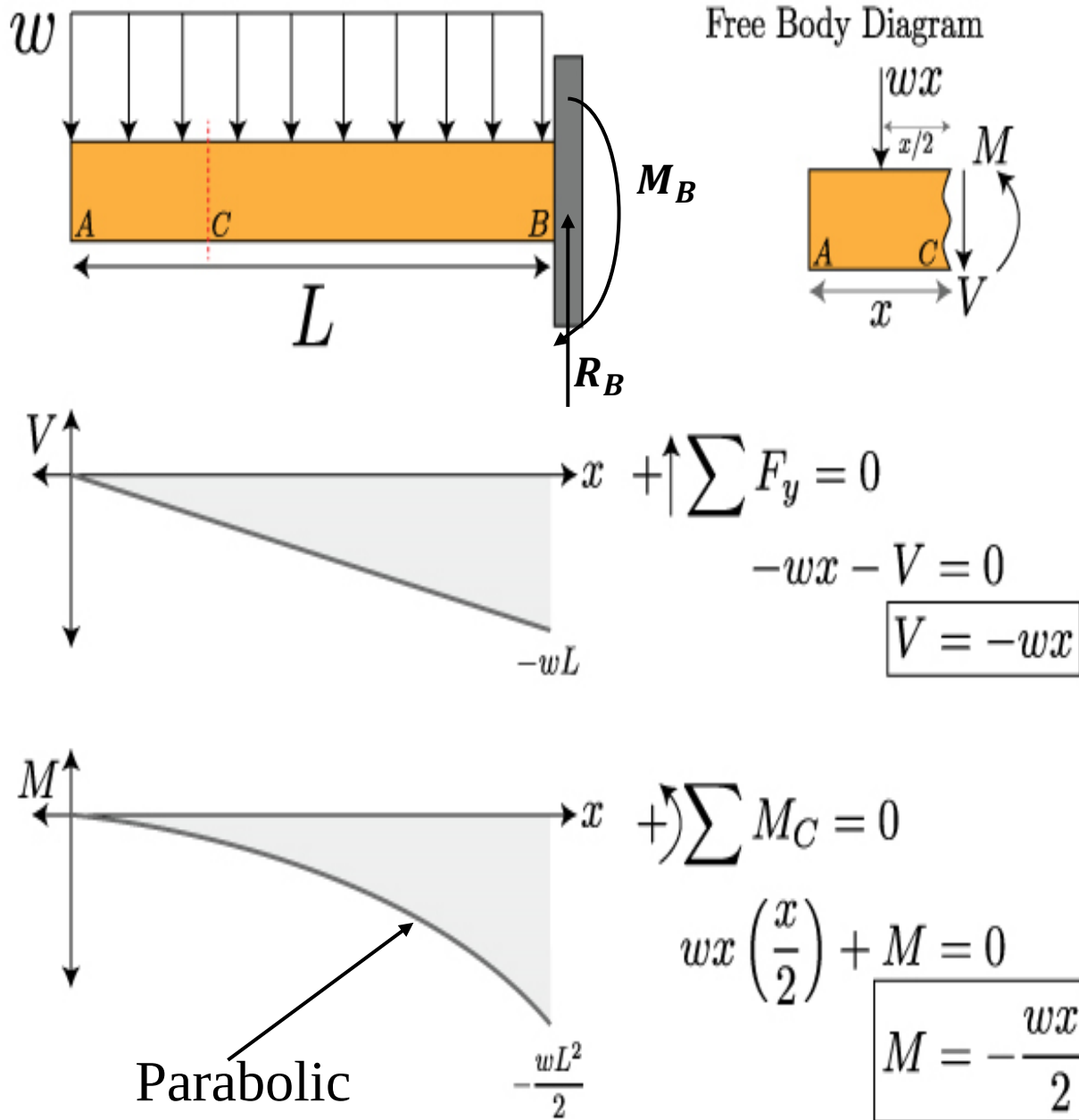


Figure (3)

□ Cantilever Beam with Uniformly Distributed Load



➤ Reactions

The reactions at A (fixed support) are obtained from equations of equilibrium.

$$R_B = wL \qquad M_B = \frac{wL^2}{2}$$

➤ Shear Force

Consider a section x-x at a distance x from free end (A).

For $0 \leq x < L$, $V_x = -wx$

Maximum shear force is at fixed end (B).

$$V_{\max} = -wL$$

➤ Bending Moment

$$M_x = -\frac{wx^2}{2}$$

Maximum bending moment is at fixed end (B).

$$M_{\max} = -\frac{wL^2}{2}$$

Cantilever Beam Carrying a Varying Load

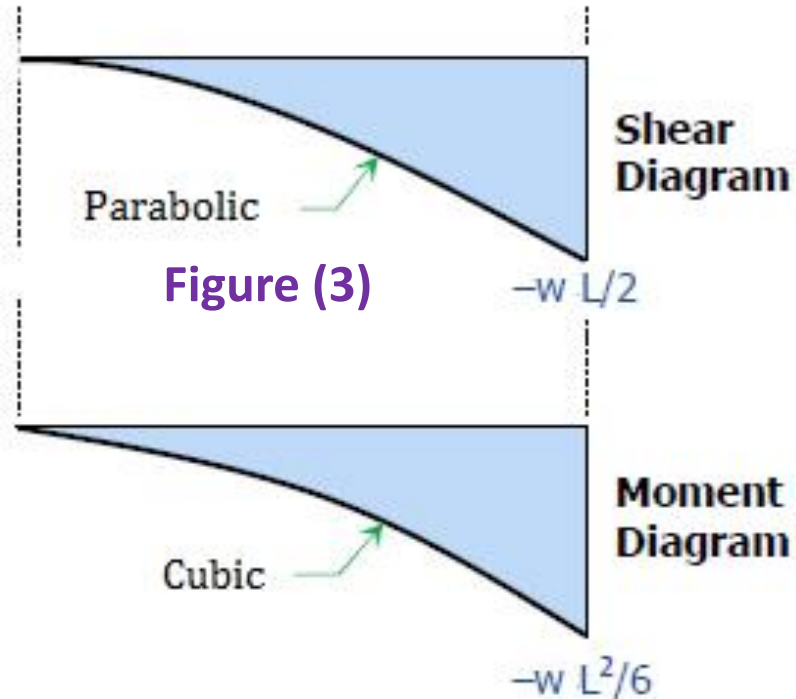
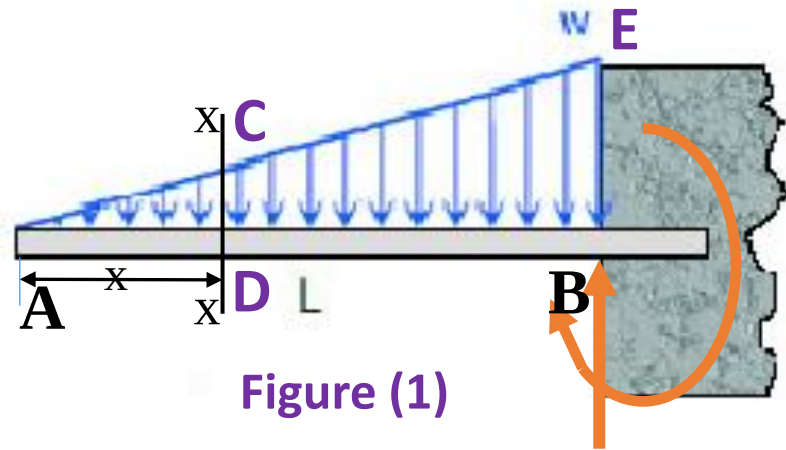


Figure (4)

Load is zero at free end (A) and **W** at fixed end (B).

➤ Reactions

$$R_B = \frac{WL}{2} \quad (i)$$

➤ Shear Force

$$V_x = -\frac{1}{2} \frac{Wx}{L} x \quad (ii)$$

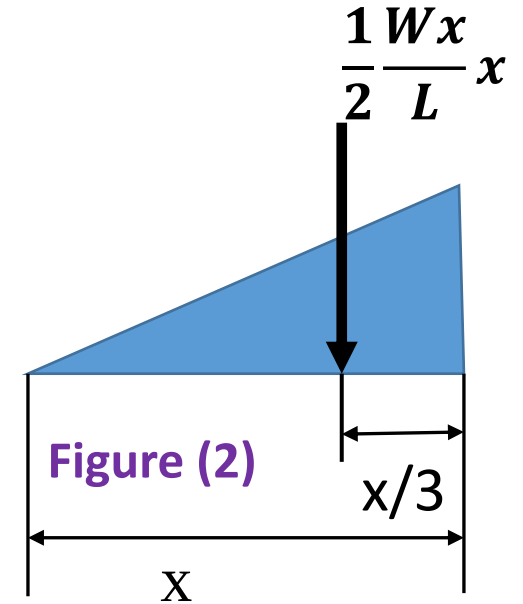
➤ Bending Moment

I. Replace the triangular loading up to x by average of it

$$\text{i.e., } \frac{1}{2} \left(0 + \frac{Wx}{L} \right) x \quad (iii)$$

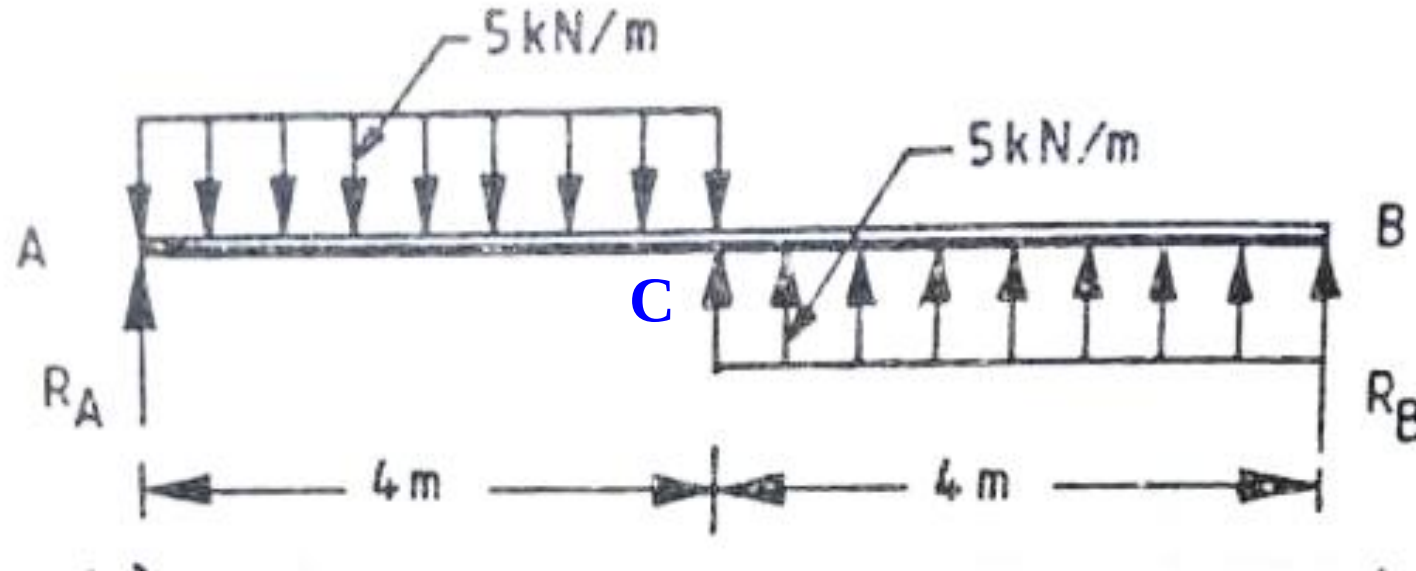
II. This average load acts through the centroid of the triangle.

III. Moment about x-x
$$M_x = - \left[\frac{1}{2} \left(\frac{Wx}{L} \right) x \right] \frac{x}{3}$$



10th Lecture

□ Draw SFD and BMD of the simply supported beam AB subjected to downward and upward distributed load.



➤ **Solution**

- Taking moment of all the forces about B, we get

$$R_A 8 - 4(5)6 + 4(5)2 = 0 \text{ (i)} \Rightarrow R_A = 10 \text{ kN}$$

- From equilibrium of forces in vertical direction, we can obtain

$$R_B = -10 \text{ kN}$$

➤ Solution

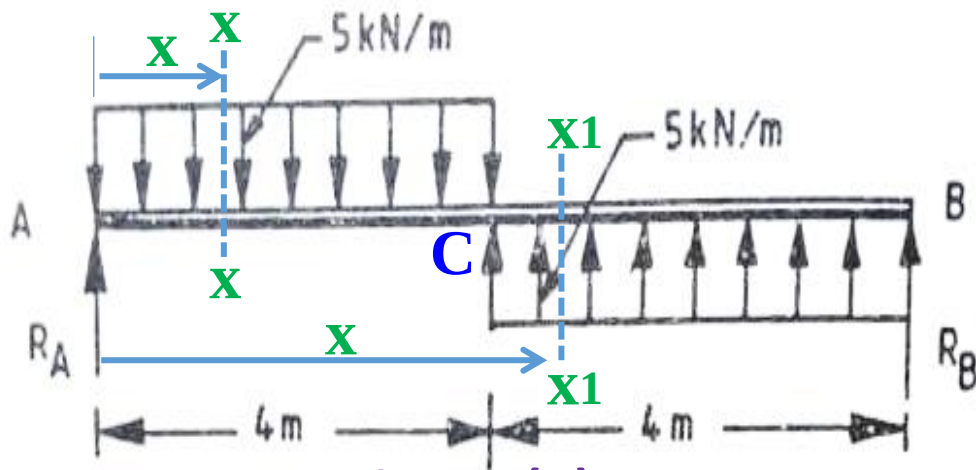


Figure (1)

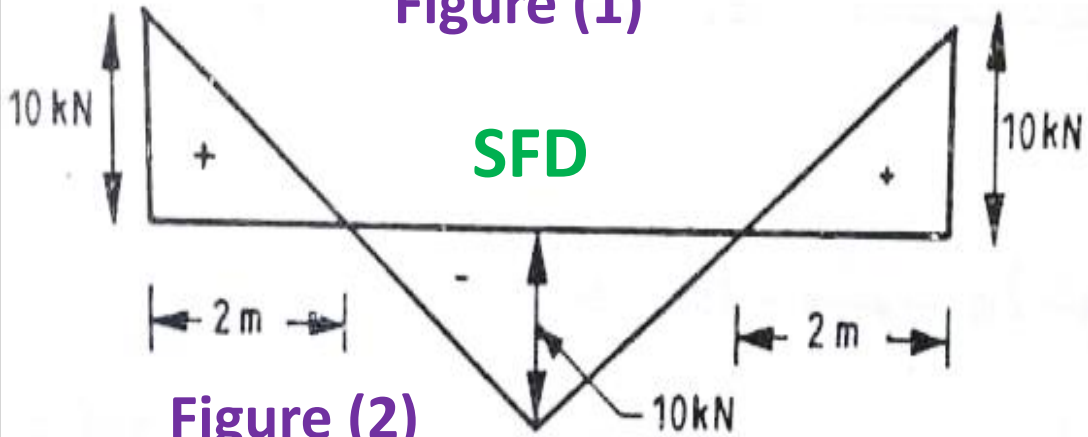


Figure (2)

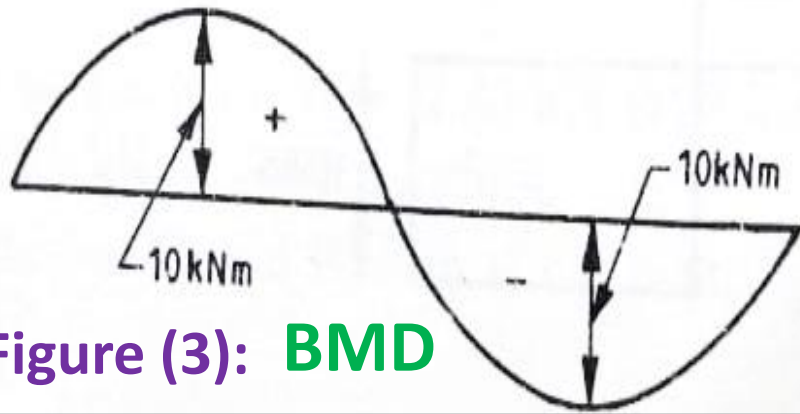


Figure (3): BMD

1. Between A and C, $0 < x < 4m$, $V_x = 10 - 5x$ (iii)

and $M_x = 10x - \frac{5}{2}x^2$ (parabolic) (iv)

2. At C, $x = 4m$, $V_c = 10 - 20 = -10$ kN (v)

and $M_c = 10(4) - \left[\frac{5(4^2)}{2}\right] = 0$ (vi)

3. Between B and C, $4m < x < 8m$,

$V_x = 10 - 20 + 5(x - 4)$ (vii)

and $M_x = 10x - 20(x - 2) + \left[\frac{5(x-4)^2}{2}\right]$ (viii)

• At B, $x = 8m$, $V_B = 0$ and $M_B = 0$. (ix)

Some Important Information

- **Notes:**

I. Where shear force changes sign in SFD, at that point bending moment is maximum or minimum in BMD.

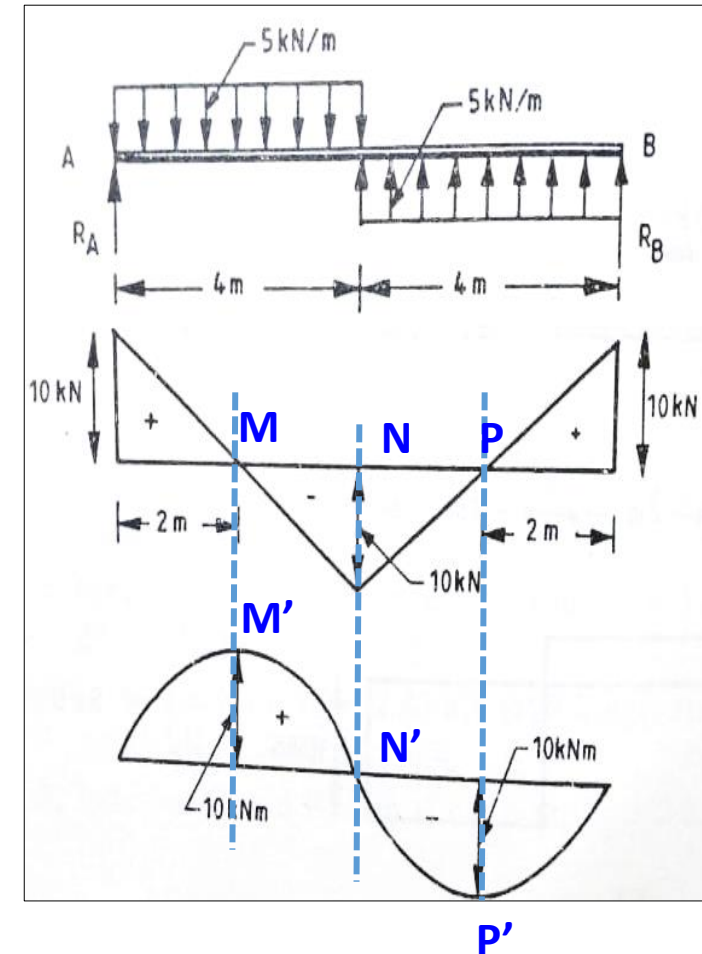
II. If bending **moment changes sign at a section**, then curvature will also change sign at that section. Such a point is called **Point of Contraflexure**.

III. The **slope of shear force curve** at any section is equal to loading rate at that section.

$$\frac{dV}{dx} = -q.$$

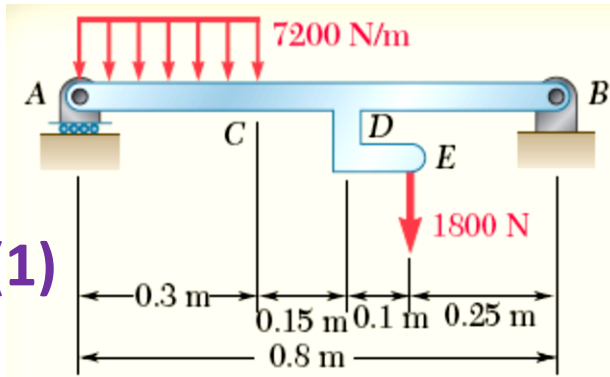
IV. **Slope of bending moment curve** at any section is magnitude of shear force at that section.

$$\frac{dM}{dx} = V$$



Sample Problem

Figure (1)



SOLUTION:

- Taking entire beam as a free-body, calculate reactions at A and B.

$$\sum M_A = 0:$$

$$B_y(0.8\text{ m}) - (2160\text{ N})(0.15\text{ m}) - (1800\text{ N})(0.55\text{ m}) = 0 \quad (\text{i})$$

$$B_y = 1642.5\text{ N}$$

$$\sum M_B = 0:$$

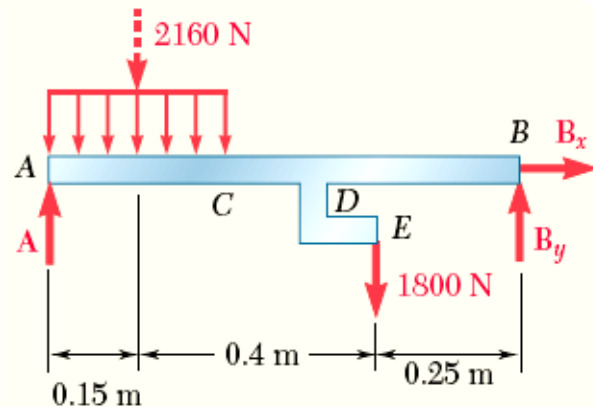
$$(2160\text{ N})(0.65\text{ m}) + (1800\text{ N})(0.25\text{ m}) - A(0.8\text{ m}) = 0 \quad (\text{ii})$$

$$A = 2317.5\text{ N}$$

$$\sum F_x = 0:$$

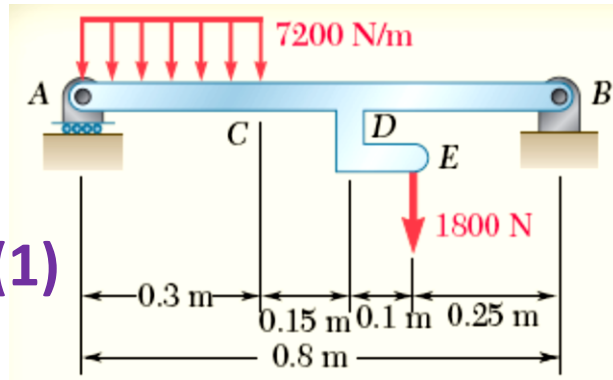
$$B_x = 0 \quad (\text{iii})$$

Figure (2)



Sample Problem

Figure (1)



- Note: The 1800-N load at *E* may be replaced by a 1800-N force and 180-N · m couple at *D*.

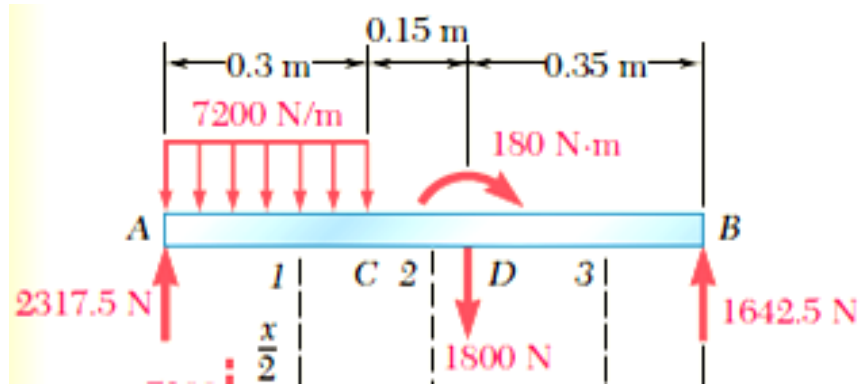
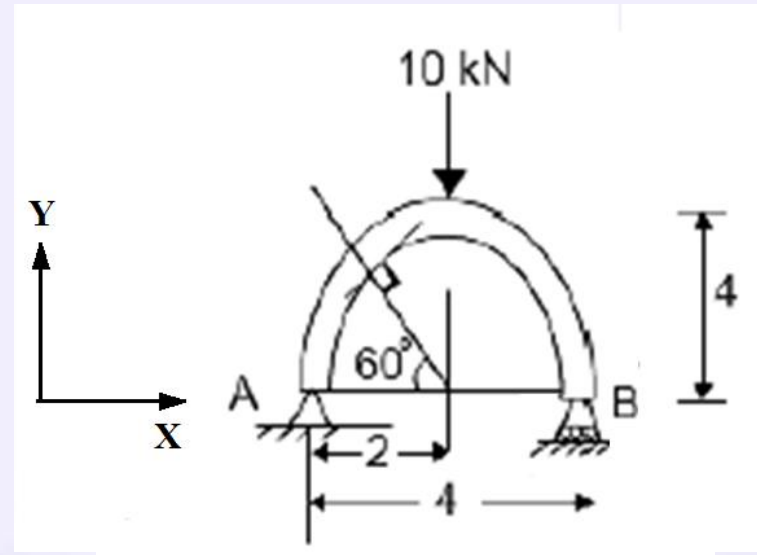


Figure (2)

□ Problem on curved beam:



A parabolic beam is subjected to a 10kN force as shown above. Find axial force, shear force and bending moment at a section, as shown above.

□ Solution on curved beam:

We first find the equation of the beam and the point where section is cut. With origin at A, equation of the parabola can be written as

$$y = kx(4 - x). \text{ It remains to find } k. \quad (\text{i})$$

$$\text{wrt } y = 4 \text{ when } x = 2 \Rightarrow k = 1$$

$$\text{so } y = x(4 - x). \quad (\text{ii})$$

$$\text{slope, } y' = \frac{dy}{dx} = 4 - 2x = \tan 30 = \frac{1}{\sqrt{3}} \quad (\text{iii})$$

$$\Rightarrow x = \frac{(4\sqrt{3} + 1)}{2\sqrt{3}} \quad (\text{iv})$$

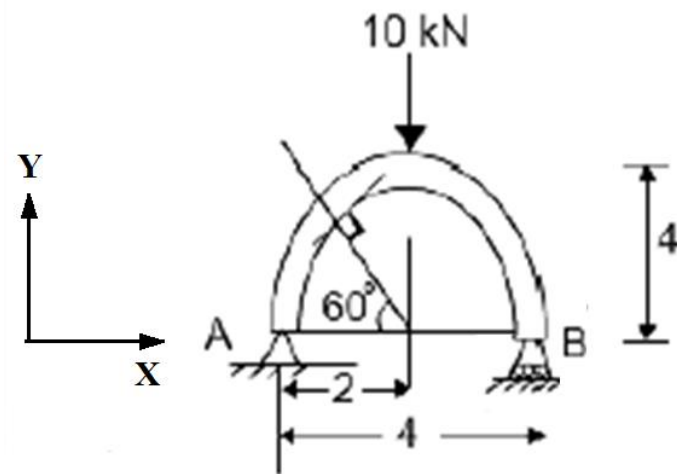


Figure (1)

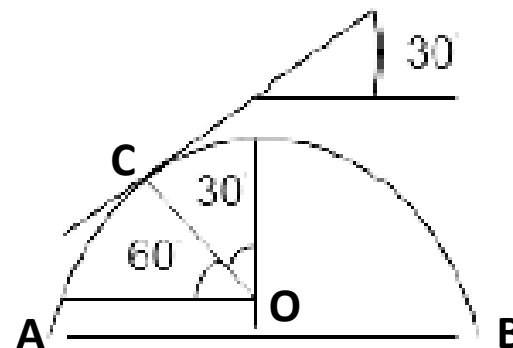


Figure (2)

Now, we only have to resolve the force at the section along the normal and tangential directions. But before that, we first have to find the support reactions.

$$\sum F_x = 0 \Rightarrow R_{Ax} = 0 \quad (\text{v})$$

$$\sum M_A = 0 \Rightarrow 10 \times 2 = R_{By} \times 4 \quad R_{By} = 5 \text{ kN}. \quad (\text{vi})$$

By symmetry (since $R_{Ax} = 0$), $R_{Ay} = 5 \text{ kN}$ [or $R_{Ay} = 10 - R_{By}$] (vii)

We now draw the free body diagram of the cut body

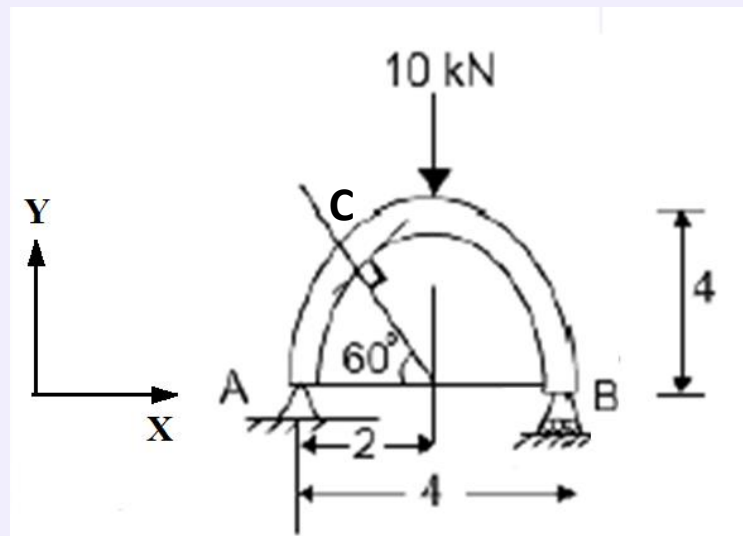


Figure (1)

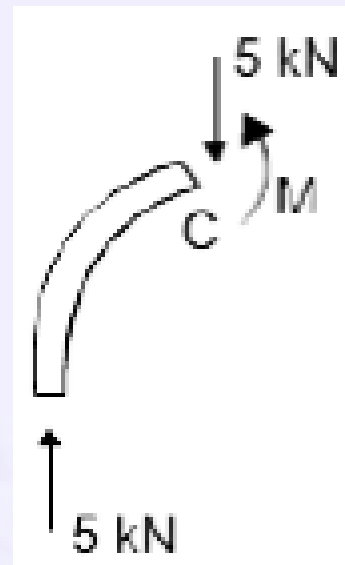


Figure (2)

We now resolve the 5kN force

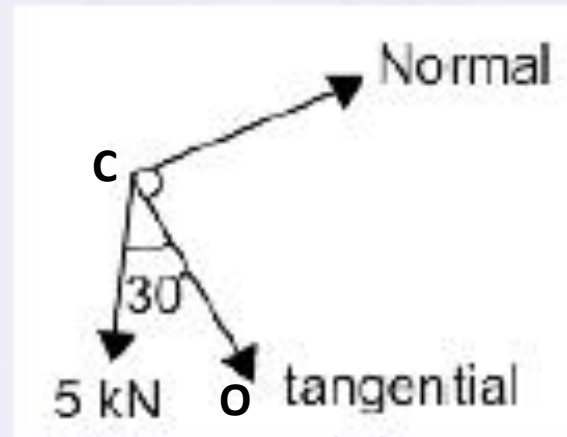


Figure (1)

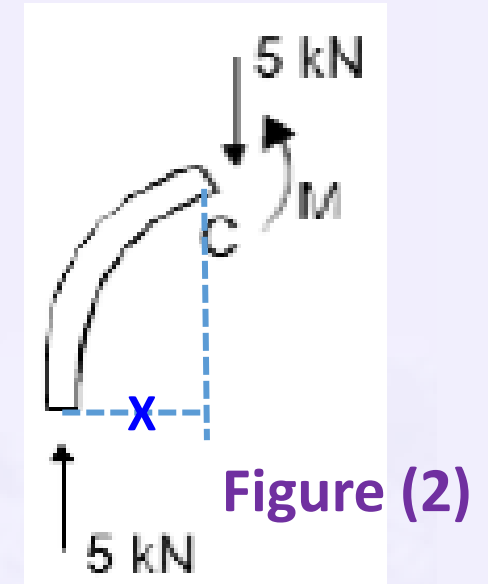


Figure (2)

$$\text{Shear force} = 5 \cos 30 = 5 \frac{\sqrt{3}}{2} = 4.33 \text{ kN} \quad (\text{i})$$

$$\text{Axial force} = -5 \sin 30 = -2.50 \text{ kN (compression)} \quad (\text{ii})$$

$$\sum M_c = 0 \Rightarrow 5 \times \frac{(4\sqrt{3}+1)}{2\sqrt{3}} = M = 11.44 \text{ kNm} \quad (\text{iii})$$

Method of Differential Equation of Equilibrium

- ✓ The derivative of the bending moment with respect to x is equal to the shear force.

$$\frac{dM}{dx} = V \quad (i)$$

- ✓ The derivative of the shear force with respect to x is equal to the load per unit length.

$$\frac{dV}{dx} = -q \quad (ii)$$

- ✓ Relation between bending moment and load,

$$\frac{d^2M}{dx^2} = -q \quad (iii)$$

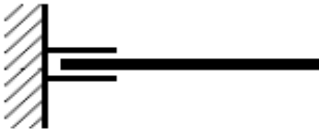
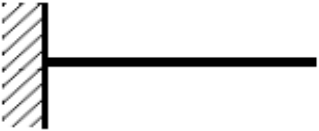
q	V	M
0	constant	linear
constant	linear	quadratic parabola
linear	quadratic parabola	cubic parabola

- The relations may also be used to quantitatively determine the values of V or M for a given load $q(x)$. If we integrate above eq. we obtain,

$$\frac{dV}{dx} = -q \quad \text{(i)} \quad V = - \int q \, dx + C_1,$$

$$\frac{dM}{dx} = V \quad \text{(ii)} \quad M = \int V \, dx + C_2.$$

- The constants of integration C_1 and C_2 can be calculated based on known values of V or M from support condition,

	support		V	M
1	pin		$\neq 0$	0
2	parallel motion		0	$\neq 0$
3	sliding sleeve		$\neq 0$	$\neq 0$
4	fixed end		$\neq 0$	$\neq 0$
5	free end		0	0

SF & BM diagrams along with Boundary Conditions of Different Loading Conditions

$$\frac{d^2 M}{dx^2} = -q$$

$$V = -q_0 x + C_1 \quad (i)$$

$$M = -\frac{1}{2} q_0 x^2 + C_1 x + C_2 \quad (ii)$$

a) $M(0) = 0,$

$M(l) = 0,$

b) $V(l) = 0,$

$M(l) = 0$

c) $V(0) = 0,$

$M(l) = 0$

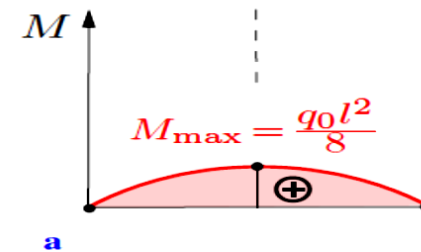
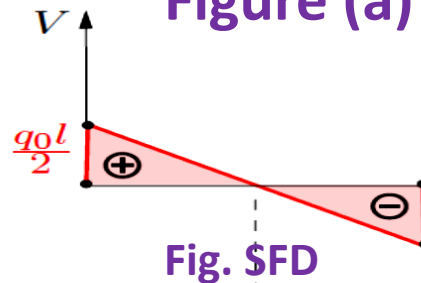
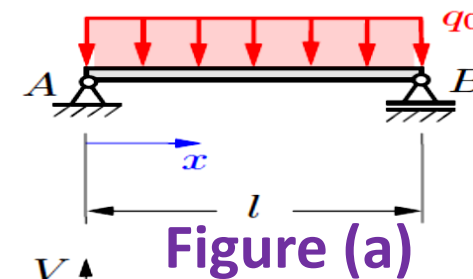


fig. 7.11

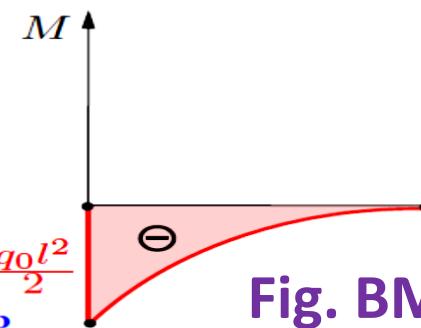
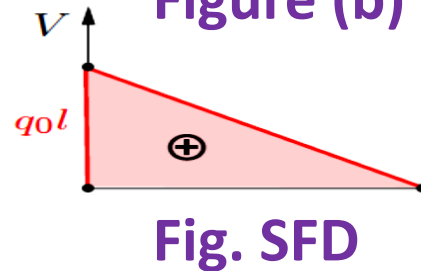
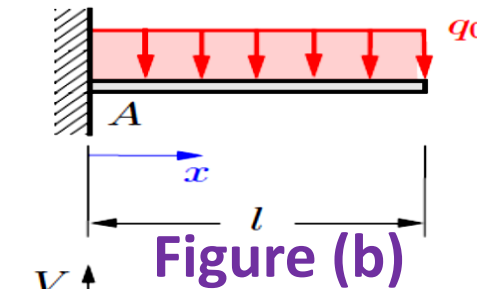
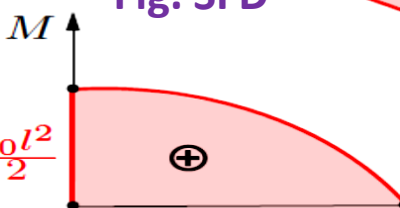
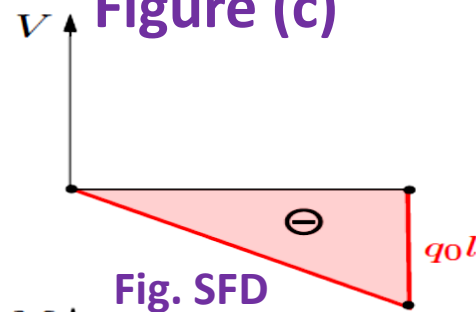
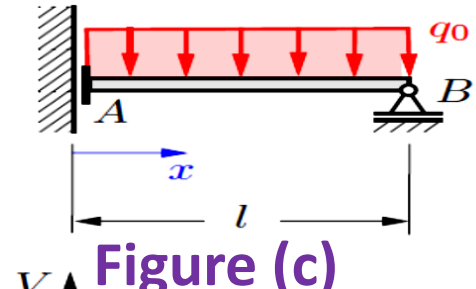


Fig. BMD



End